

# CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Second Semester of M. Tech. Examination (C.E.)

May 2018

CE765 Machine Learning (ML) (Program Elective-II)

Date: 08.05.2018, Tuesday Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

## Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Rough work is to be done in the last page of main supplementary, please don't write anything on the question paper.
5. Indicate clearly, the option(s) you attempt along with its respective question no.
6. Figures to the right indicate marks.

## SECTION-I

### Q-1 Answer the following questions.

1. Explain the concept learning in brief. List down the application areas where concept learning works better than other approaches. 4
2. What is Machine Learning? What is the need of it? Briefly explain the Machine Learning characteristics. 2
3. What is Classification? Which are the phases of Classification process? Differentiate between Binary Classification and Multiclass Classification. 5

### Q-2

- [A] How to measure the performance of a Classifier? How to compare the performance of one Classifier with the other? What is a Confusion Matrix? What is a Contingency Table? 4

OR

- [A] Explain the Regression along with an equation. How to measure performance of Regression? Describe error calculation in Regression. 4
- [B] Differentiate between Univariate Regression and Multivariate Regression. 4
- [C] What is over-fitting and under-fitting? Which are the reasons for over-fitting and under-fitting? 4

OR

- [C] Explain the k-NN classification along with example. Which are the effects of k on result of k-NN classifier? 4

### Q-3

- [A] Explain the parameter K in K-Means Clustering Algorithm. Explain K-Means Clustering Algorithm along with an example. 4

OR

- [A] Explain the role played by the Bayes' Theorem in Naïve Bayes classifier. What is Naïve Bayesian Classifier? Explain prior Probability, Posterior Probability and Maximum Likelihood with respect to Naïve Bayesian Classifier. 4
- [B] What is Perceptron? What is Multi-Layer Feed Forward Networks? What is SVM? How does SVM differ from Artificial Neural Networks (ANN)? 4
- [C] Define and explain in brief about the various input parameters to be set for GA (Genetic Algorithm). Differentiate between Steady-State GA and Generational GA. 4

**OR**

- [C] Discuss the BP (Back Propagation) NN (Neural Network) learning algorithm along with its mathematical derivation. Discuss briefly about the limitations of Back Propagation Learning Algorithm. 4

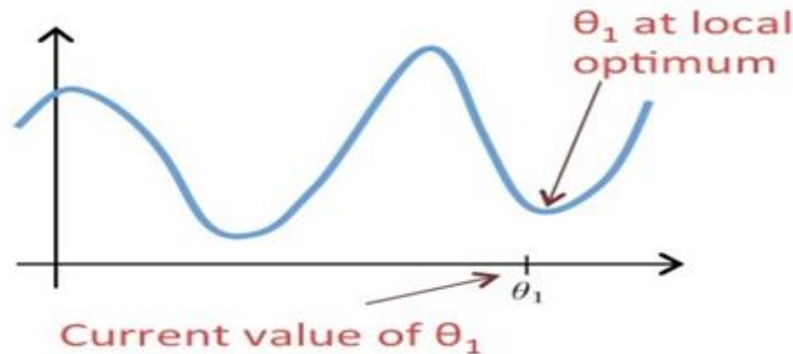
**SECTION-II**

**Q-4 Answer the following questions.**

1. Explain Supervised, Unsupervised and Reinforcement learning using an example. 4
2. Give three computer applications for which Machine Learning approaches seem appropriate and three applications for which they seem inappropriate. 3
3. Explain in brief Descriptive and Predictive Tasks of Machine Learning. 4

**Q-5**

- [A] Why are crossover and mutation required? Define Crossover Rate and Mutation Rate. Explain Roulette Wheel Selection with an example. 5
- [B] Let  $X=\{a,b,c,d\}$ ;  $Y=\{1,2,3,4\}$  be two universe of discourse.  $A=\{(a,0),(b,0.8),(c,0.6),(d,1)\}$ ,  $B=\{(1,0.2),(2,1),(3,0.8),(4,0)\}$  and  $C=\{(1,0),(2,0.4),(3,1),(4,0.8)\}$ . Prove that any one De Morgan's Law and determine the implication relations: IF  $x$  is  $A$  THEN  $y$  is  $B$  ELSE  $y$  is  $C$ . 4
- [C] Suppose  $\theta_1$  is at local optimum of  $J(\theta_1)$  such as shown in the figure. What will one step of gradient descent will do? Justify your answer. 3



**OR**

**Q-5**

- [A] What is significance of Hybrid System in Soft Computing? Explain pros and cons of various Hybrid Systems in Soft Computing using GA, NN and FS. 5
- [B] What is Defuzzification? Explain at least one method with its equation and also explain it with suitable example. 4
- [C] What are the benefits of Multiple Classifier System (MCS) in comparison with Single Classifier System? 3

**Q-6**

- [A] What is the effect of learning rate  $\alpha$  in the performance of gradient descent? Also justify the statement: "Gradient Descent can convergence to a local minimum, with the learning rate  $\alpha$  is fixed." 4

**OR**

- [A] Define Encoding in GA. Write the Encoding representation of Transportation Problem. 4

- [B] Why are there less parametrs in Convolutional Neural Networks in comparation with Artificial Neural Networks? 4

**OR**

- [B] Let  $A=\{(x1,0.2),(x2,0.7),(x3,0.4)\}$  and  $B=\{(y1,0.5),(y2,0.6)\}$ ,  $C=\{(z1,0.3),(z2,0.5),(z3,0.6)\}$  be three fuzzy sets defined on the universe of discourse X, Y and Z respectively. Generate relations R, S resulting out of the fuzzy Cartesian product  $A \times B$  and  $B \times C$  respectively. After that yield  $R \circ S$  by max-min composition. 4

- [C] Why is padding considered as one of the important building block of Convolution Neural Network? 4

**OR**

- [C] Explain general block diagram of Fuzzy Inference System with an example. 4